



From Chaos to Control

Navigating Airflow Sprawl with Centralized Observability

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Your Presenter



Jon Hiett

Global Solution Architect
Broadcom

~25 Years
Automation Experience

Expertise in **Financial**
and **IT** sectors

Based in the **United**
Kingdom

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The Airflow Paradox

"Airflow's greatest strength—its ease of localized deployment—has become the enterprise's greatest vulnerability."

We traded centralized control for developer velocity, creating isolated data silos that mask systemic risks and erode engineering productivity.



The Reality of "Airflow Sprawl"



Fragmented Ecosystems

Organic growth leads to multiple isolated instances across teams. This empowers localized speed but destroys global visibility.



Hidden Security Risks

Sprawl often results in key datasets being managed by outdated, insecure, or unpatched setups operating completely off the grid.



Broken Data Lineage

When one team's DAG serves as the input for another team's workflow on a different instance, end-to-end tracing becomes impossible.



The Developer Experience



The Local Dev Illusion

DAGs run perfectly in development, but break in production when interacting with hidden cross-team dependencies across a sprawling architecture.



Fragmented Debugging

Troubleshooting often requires jumping across disparate UI's and manually stitching together disconnected logs just to find a root cause.



Context-Switching Fatigue

Engineers are forced into detective work, burning cycles on reactive root-cause analysis instead of innovating.



The True Cost of Blind Spots

75%

Investigation & Context Gathering

25%

Actual Resolution & Remediation

In distributed systems, the vast majority of incident time is wasted simply trying to figure out where and why a failure occurred.

Source: Sherlocks AI (2026). "How to Reduce MTTR in Distributed



The Limitations of Native Observability

"Green" doesn't mean successful.

Native Airflow provides excellent execution status for isolated tasks, but it lacks business context.

- ▶ It cannot tell you if an upstream enterprise scheduler delivered a corrupt file.
- ▶ It cannot predict if a delay in DAG A will cause a financial SLA breach in DAG B on a different instance.
- ▶ Engineers are forced into reactive firefighting, building brittle custom monitoring tools.



Regaining Control

Moving from reactive firefighting to proactive, business-aware data operations.



Automation Analytics & Intelligence (AAI)

Unified Observability

AAI acts as a vendor-agnostic intelligence layer.

It connects your entire Airflow landscape—across all instances, versions, and even legacy workload automation platforms—into a single pane of glass.

Business-Aware Management

Move beyond asking, "Is my task green?"

AAI shifts the paradigm to asking, "What is the business impact if this dataset is late?" by tracking SLAs based on actual business outcomes.



Predictive SLA Management

24/7

Continuous Forecasting

Anticipating the Breach

Using historical data, statistical modeling, and real-time processing loads, AAI dynamically forecasts SLA outcomes.

It identifies emerging delays and issues proactive alerts well before business service delivery is actually impacted, allowing teams to intervene early.

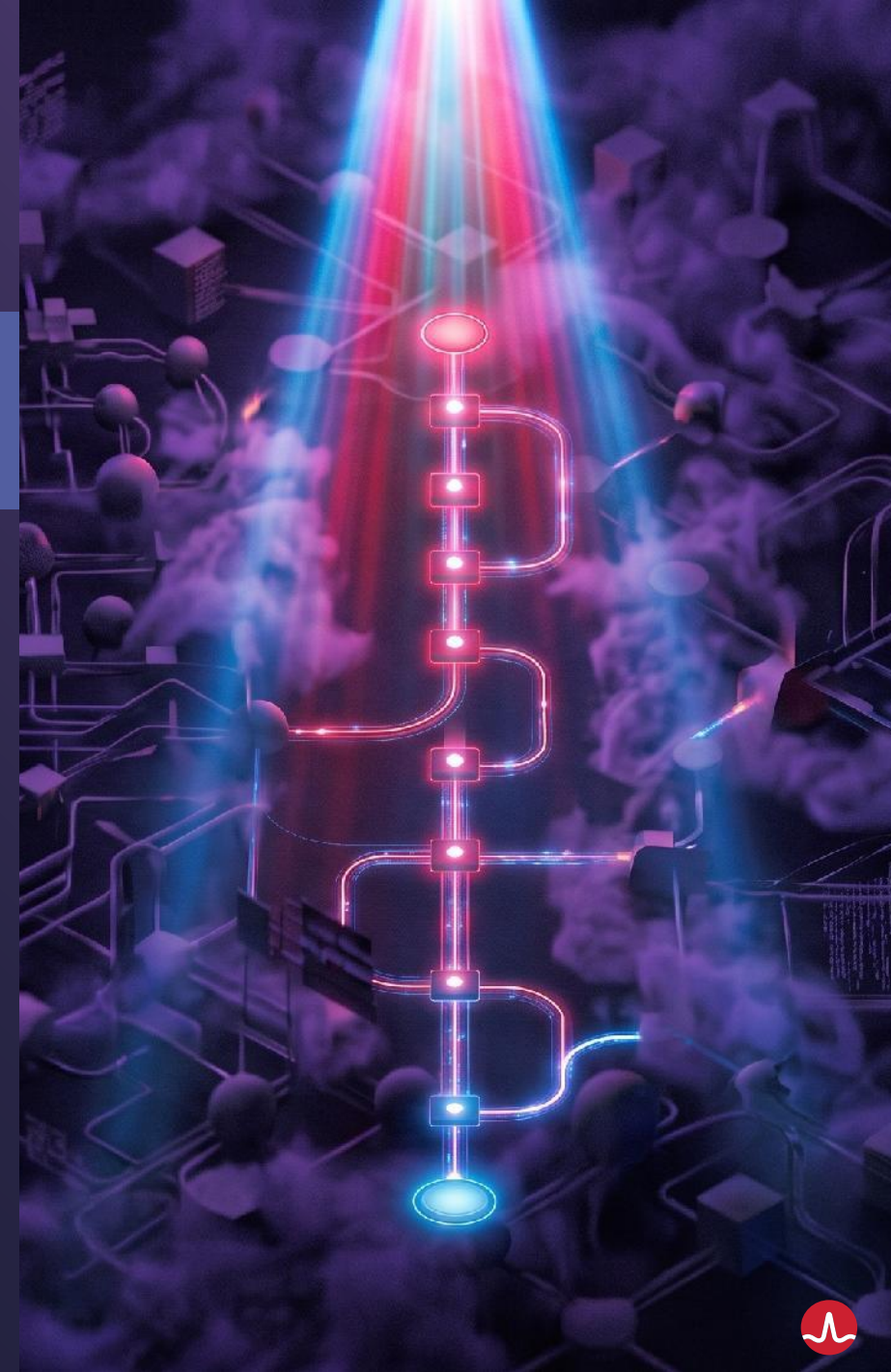


Dynamic Critical Path Discovery

Illuminating the Blind Spots

Understanding dependencies across a fragmented architecture is nearly impossible manually.

- ▶ AAI instantly visualizes the entire business process.
- ▶ It dynamically maps the critical path across disparate Airflow deployments.
- ▶ This allows DataOps teams to pinpoint the exact root cause of a failure, even if it originates far outside their specific Airflow cluster.



Empowering the Developer Workflow

Code-to-Production Confidence

Developers are tired of pipeline changes breaking downstream dependencies they couldn't see.

"Shift-Left" Reliability

AAI provides developers with global dependency visibility *before* code is deployed.

Reduced Alert Fatigue

Stop calling engineers for isolated task failures that don't actually impact business SLAs.

Contextual Alerting

Route alerts based on predicted SLA breaches, not just rigid state changes.



Proving Business Value



The Value Gap

Data teams build critical workflows, yet are often viewed as cost centers.

Translating technical pipeline execution into demonstrable business ROI has historically been an uphill battle.



Data FinOps & Chargeback

AAI V26 introduces comprehensive chargeback visualization.

Map Airflow pipeline executions and costs directly to business units, quantifying exactly who relies on your data products.



GenAI Reporting

Leverage Generative AI to instantly construct custom dashboards.

Ask plain-language questions to surface pipeline insights without manual reporting.



A Strategic Framework for Scale



Centralize Visibility, Not Just Execution

You don't have to force teams onto one Airflow cluster. Let them move fast locally but centralize the telemetry and observability globally.



Define Business-Driven SLAs

Stop measuring success by "task completion time." Measure success by the reliable delivery of data products to the business stakeholders.



Integrate Predictive Analytics

Move away from static threshold alerts. Use historical modeling to alert on future breaches, not past failures.



Bridge the Automation Silos

Recognize that Airflow is just one part of the enterprise fabric. Connect it with your other automation tools to secure end-to-end lineage.





You can't scale what you can't see.

Are your data pipelines a cohesive service,
or just a collection of isolated scripts?

Thank you for your time | Q & A



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